Jinyoon Kim

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Website: research & project details | GitHub | LinkedIn

Education

University of Virginia *M.S. Computer Science, 2025–2027*
Cumulative GPA: **3.963/4.00**
Pennsylvania State University *B.S. Computer Science, 2024*

Research Interests

Embodied AI, part-guided manipulation, human–robot interaction, and visual understanding.

Current Research

Part-Guided Robotic Manipulation (Part2Action) [GitHub]
Advised by Prof. Yen-Ling Kuo. Study how robots follow instructions about object parts to guide manipulation, with the goal of extending this work to human–robot interaction and assistance.

Long-Horizon Interactive Robotics [GitHub]
Contribute to UVA Link Lab research linking navigation and physical interaction over longer task sequences, exploring robot foundation models for reliable interaction.

Selected Publications

J. Kim, et al. *Automated Image Segmentation Using Self-Iterative Training and Self-Supervised Learning with Uncertainty Scores.* Book chapter, Chapman & Hall/CRC, 2025. [GitHub]

J. Kim, et al. *YOLO-SCSA: Enhanced YOLOv8 with Spatially Coordinated Shuffling Attention Mechanisms for Skin Cancer Detection.* ICMLA, 2024. [GitHub]

Selected Projects

Part-Aware Policy Learning *(ongoing)* [GitHub]

- Investigate whether task-relevant object-part representations improve imitation-learning policies for robotic manipulation and generalization to unseen objects.

Medical Drone Landing-Pad Coordination (SMGLib) [GitHub]

- Implemented priority-based coordination, holding orbits, negotiation, round-trip deliveries, and baseline trajectory planning; integrated collaborators' lookahead and LLM methods.

Text-Guided 3D Scene Editing [GitHub]

- Built a 3D Gaussian Splatting pipeline for text-guided object removal and addition, combining occlusion-aware object selection, multi-view inpainting, and generative 3D assets.

BGP Routing Threats and Route-Leak Detection [GitHub]

- Built a BGP analysis pipeline to evaluate ASPA path validation and route-origin validation through historical incident replay and deployment-coverage studies.

Reinforcement Learning for Financial Reasoning [GitHub]

- Developed an RL fine-tuning and evaluation pipeline for financial question answering, comparing policy optimization methods and reranking strategies across language models.

Technical Skills

Languages: Python, C++, Java, LaTeX **Tools:** PyTorch, OpenCV, Docker, Linux, Git, AWS

Experience & Awards

Intern, K&C Love Consulting Corp. (2024–2025). Supported ML data preparation and documentation.
Awards: Ackroyd Healthier Days Scholarship (2024); Penn State CS Undergraduate Award (2023).